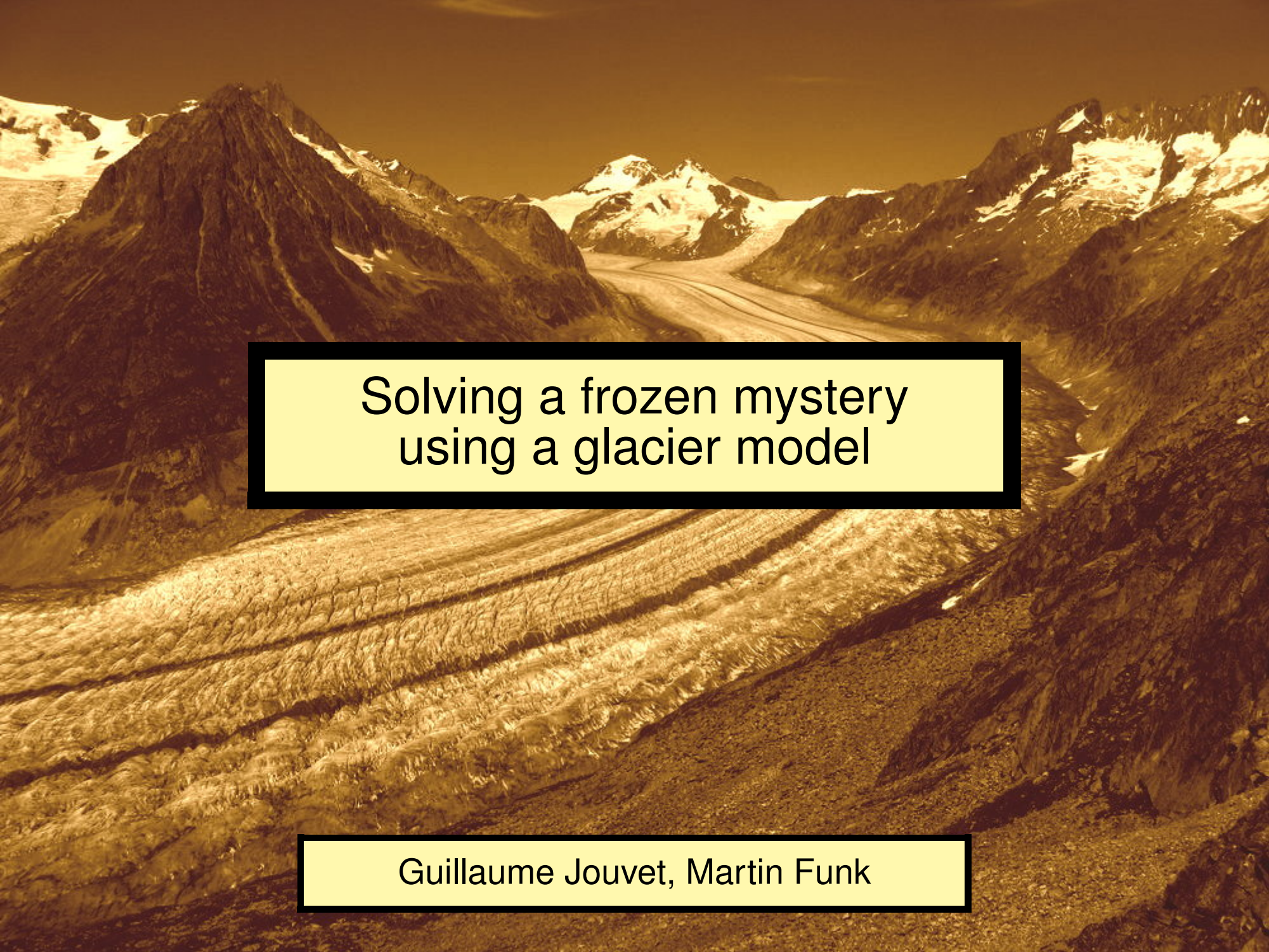
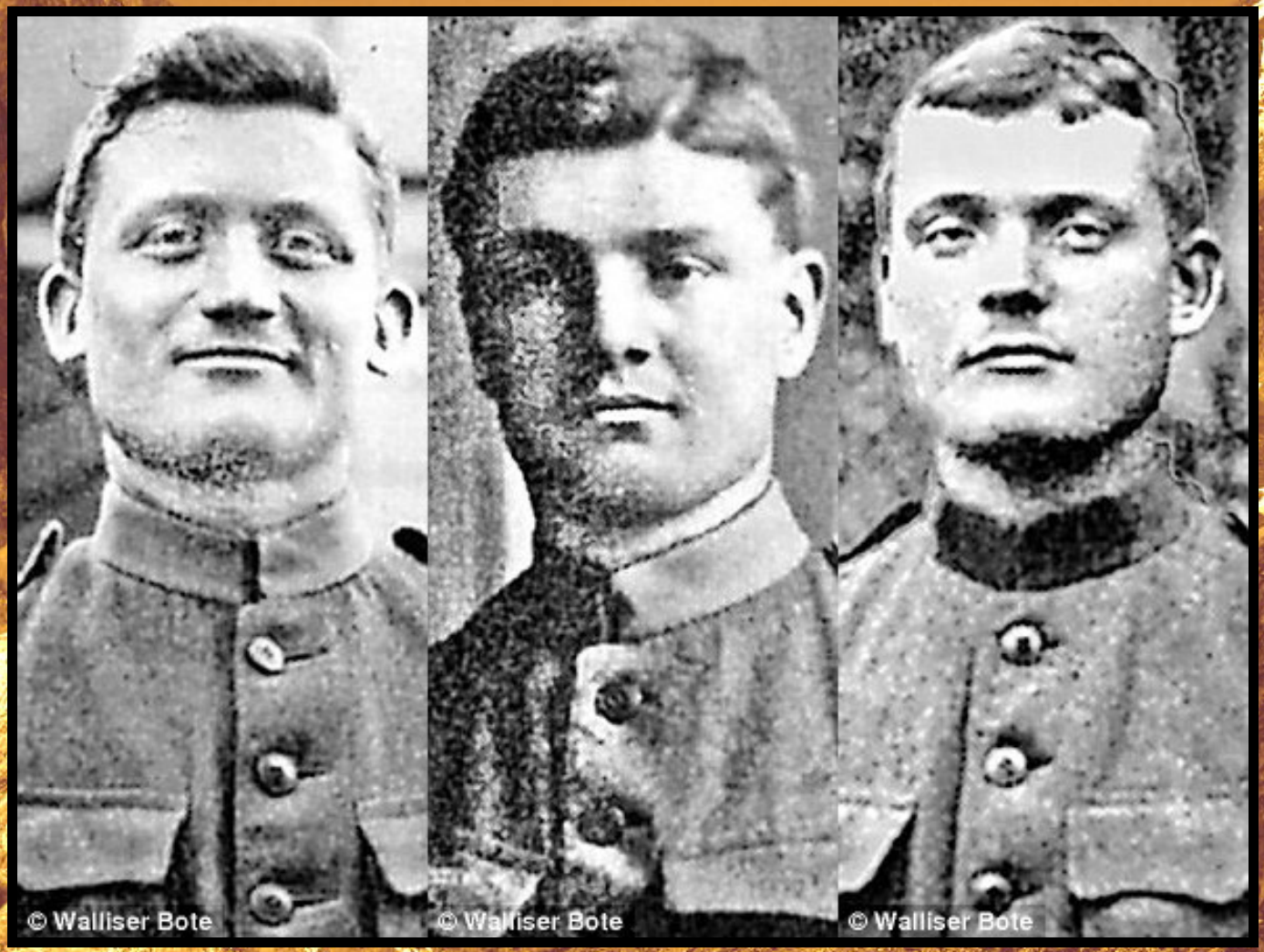


Solving a frozen mystery using a glacier model

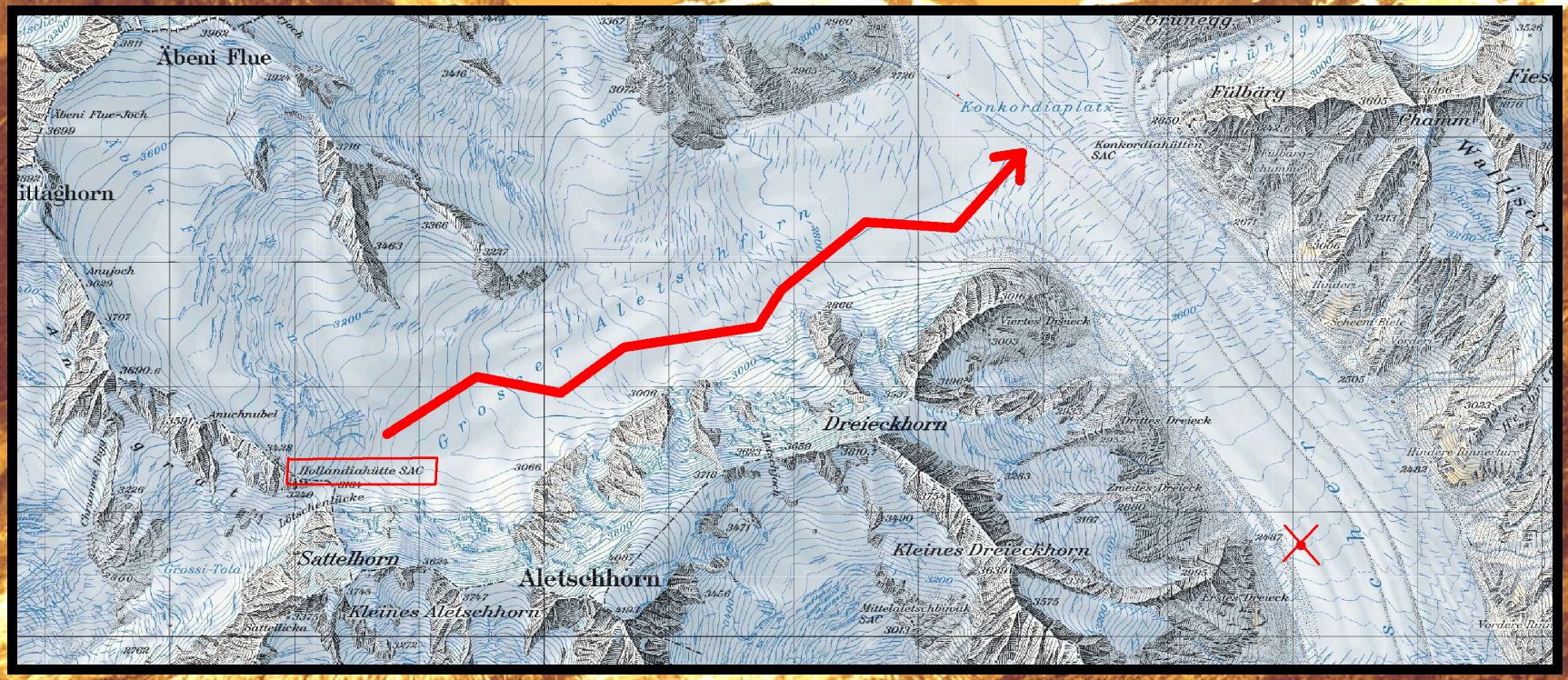
Guillaume Jouvet, Martin Funk

Ebener's brothers story (1926)



Fidelis, Cletus and Johann Ebener

Ebener's brothers story (1926)



4/3/1926 at noon, they left the refuge to go to **Konkordiaplatz**.

Ebener's brothers story (1926)



A **snow-storm** broke out during the afternoon.

Ebener's brothers story (1926)



Despite search operations, **no trace** of the group could be found

Ebener's brothers story (1926)



Their remains were found in 2012.

Summer 2012

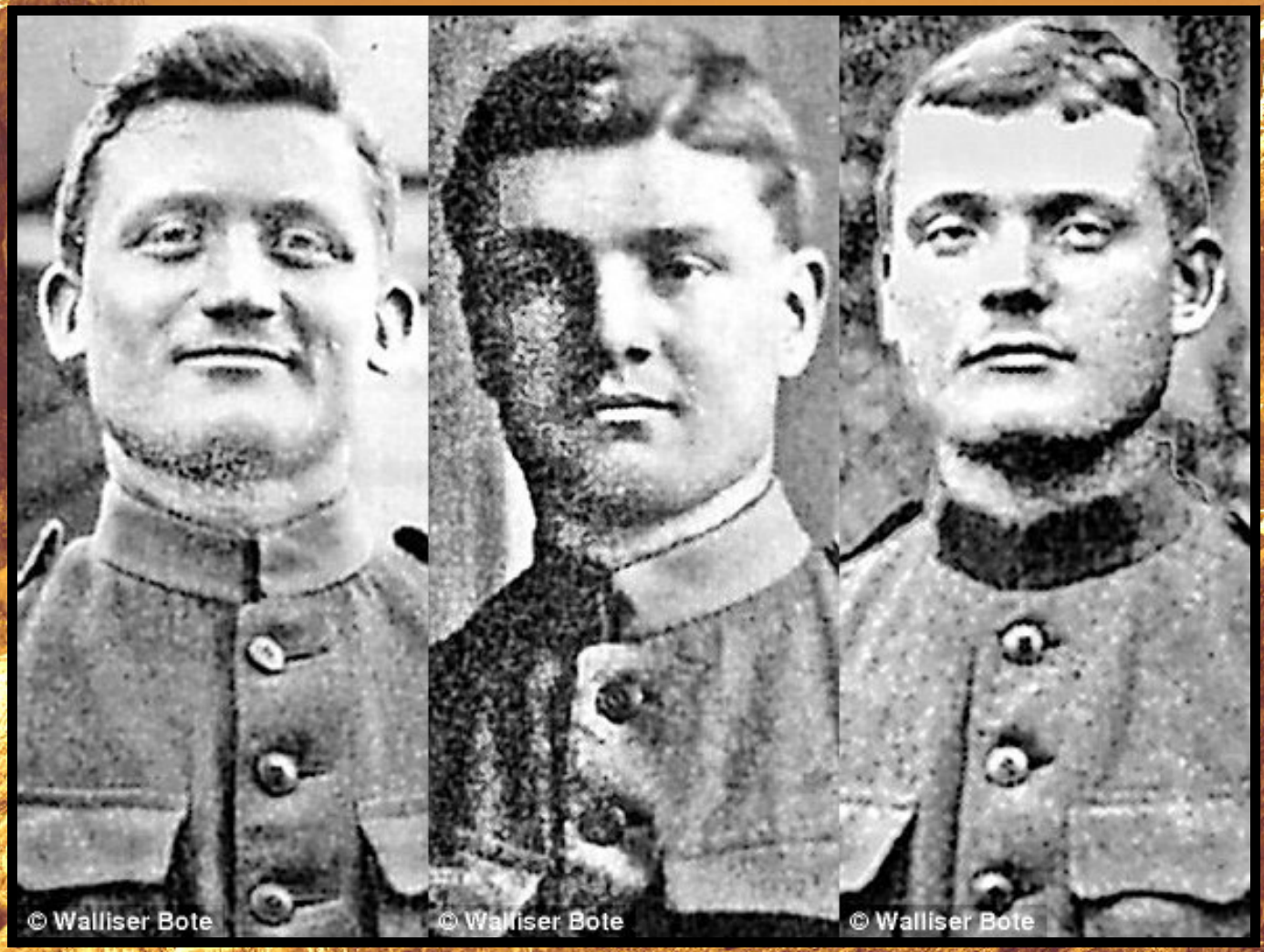








Summer 2012



DNA analysis confirmed remains belong to Ebener's brothers

Where the bodies have been buried?



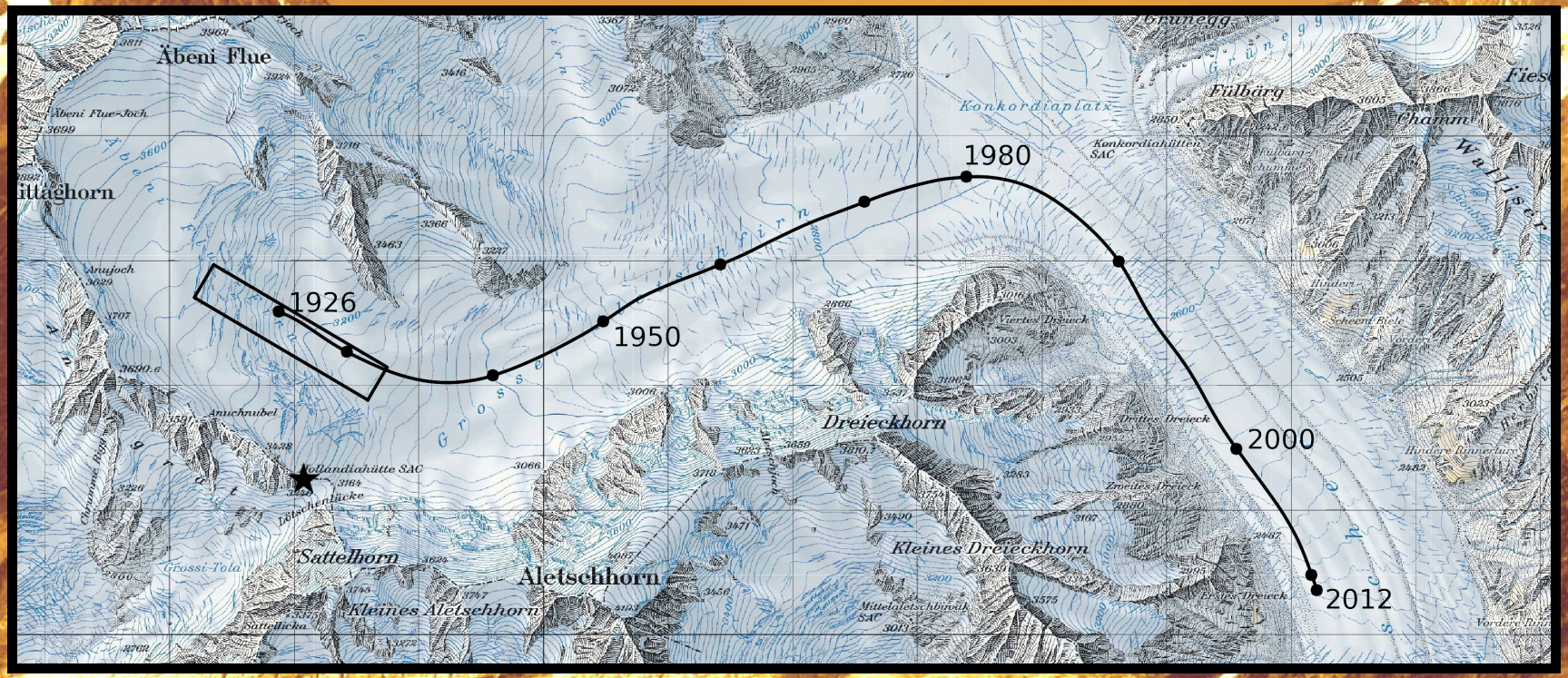
Where the bodies have been buried?



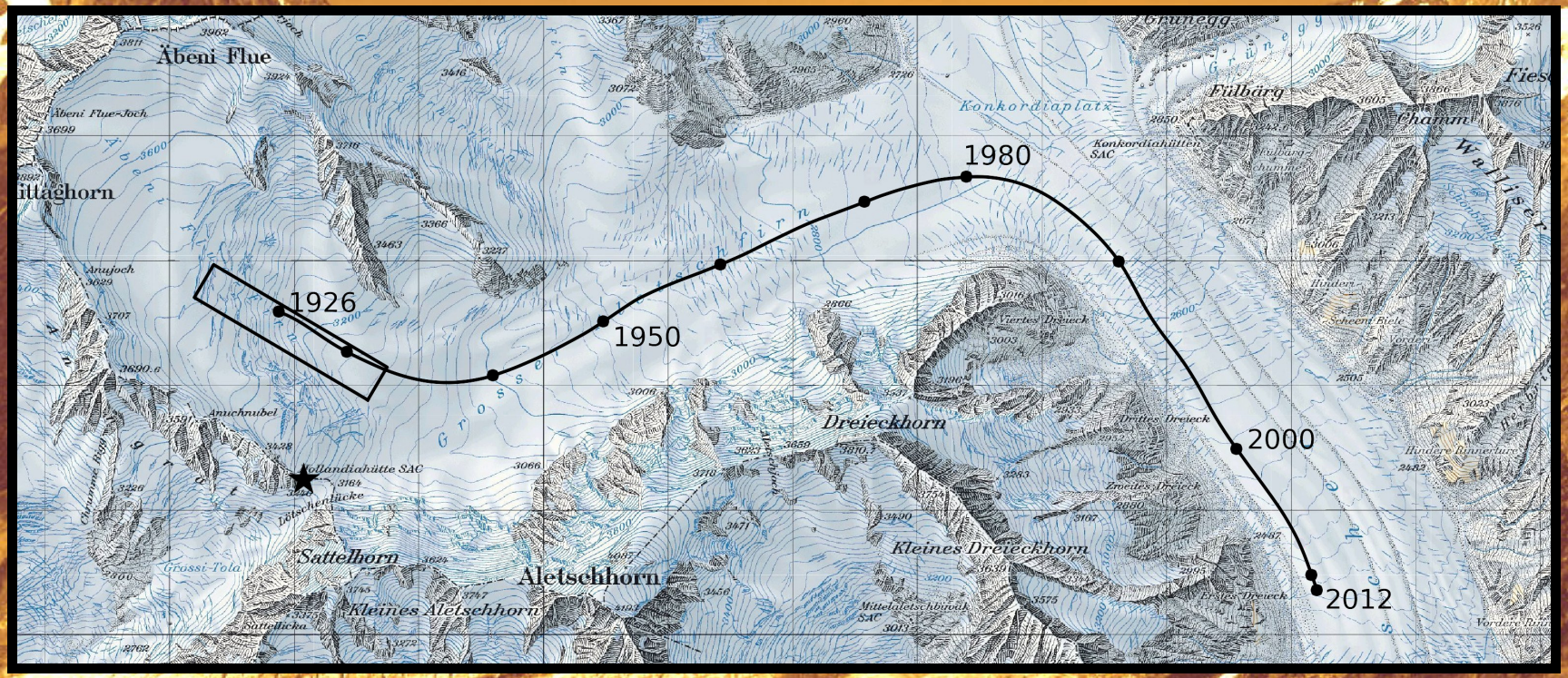
The glacier modelling can give evidences!

⇒ *Animation 1*
⇒ *Animation 2*

Trajectory of the corpses in the glacier

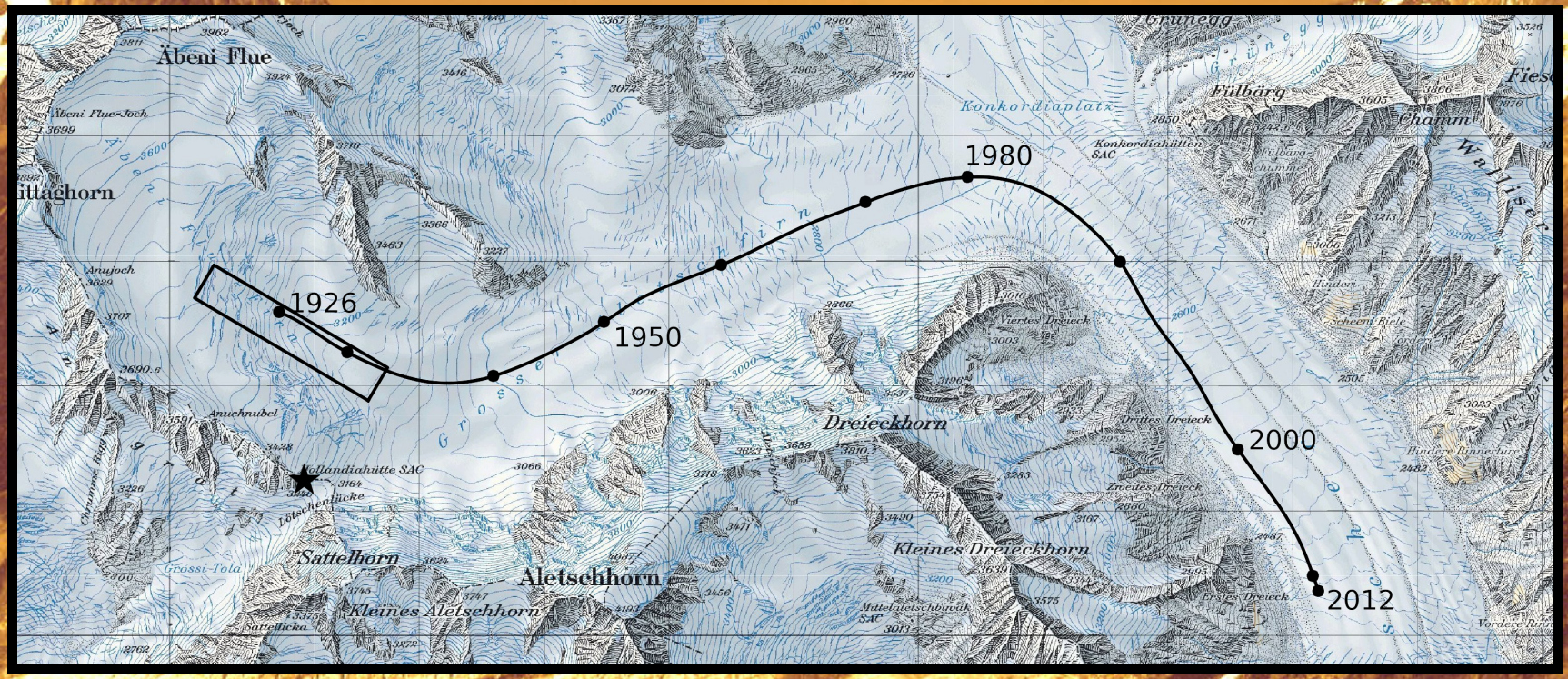


Trajectory of the corpses in the glacier



Three causes of death are conceivable:

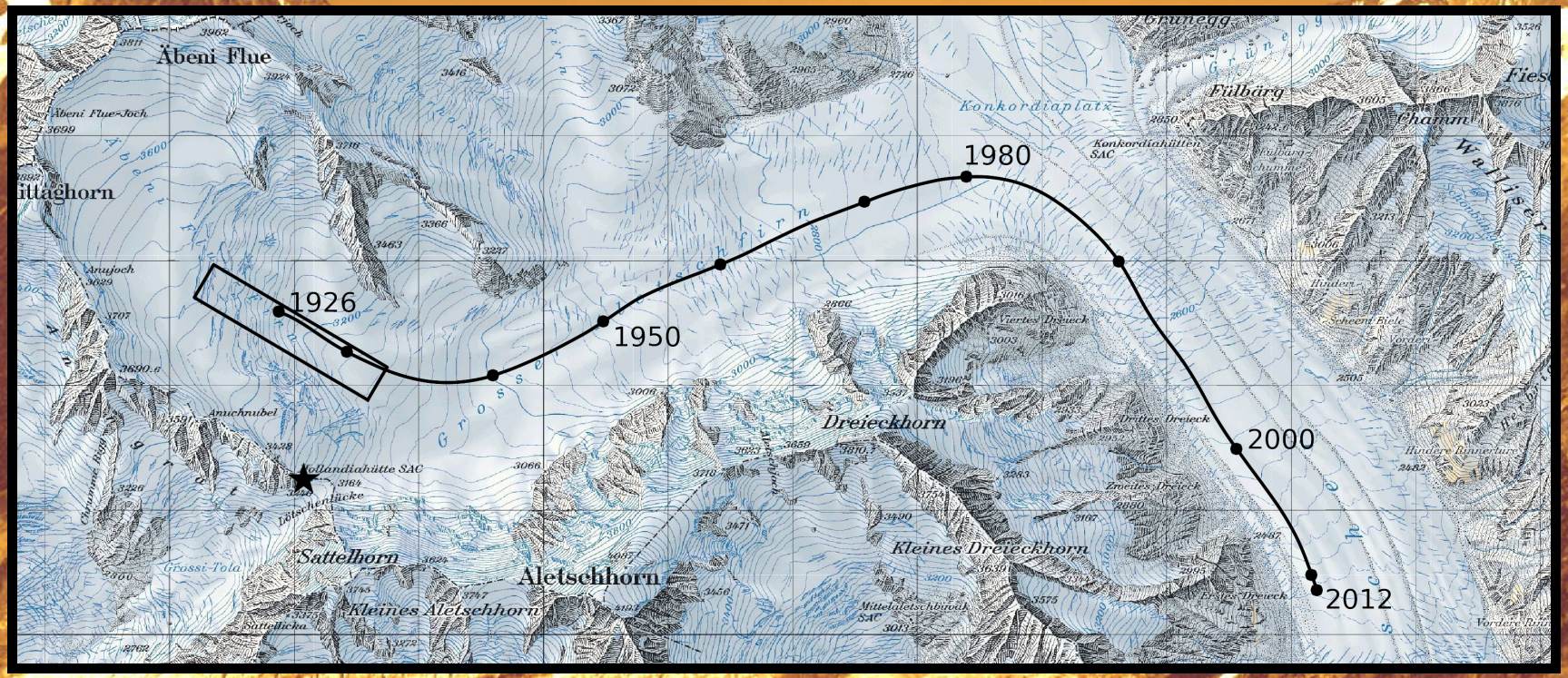
Trajectory of the corpses in the glacier



Three causes of death are conceivable:

- (1) fall into a crevasse

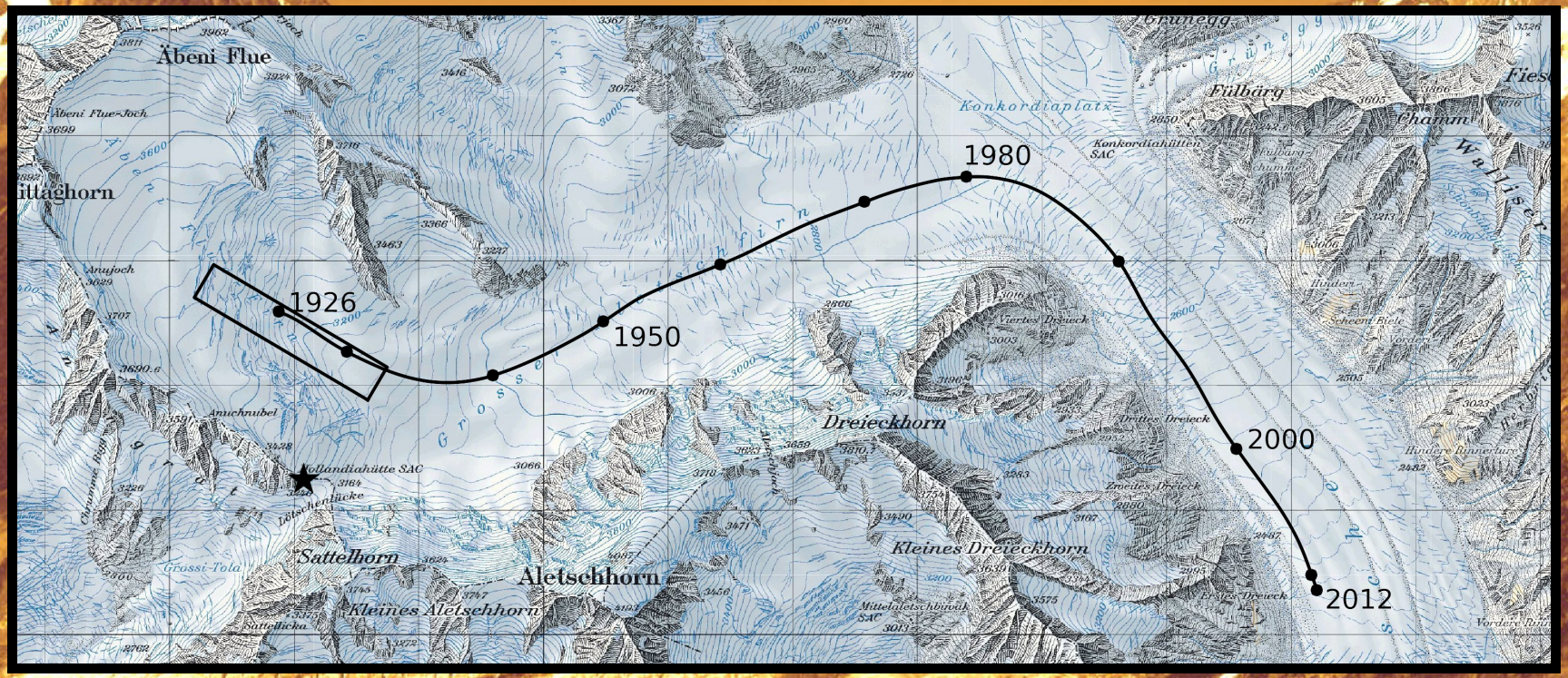
Trajectory of the corpses in the glacier



Three causes of death are conceivable:

- (1) fall into a crevasse
- (2) avalanche burial

Trajectory of the corpses in the glacier



Three causes of death are conceivable:

- (1) fall into a crevasse
- (2) avalanche burial
- **(3) exhaustion/freezing to death**

Gauligletscher: Another application?

1946: A US aircraft crash-lands on the Gauli glacier, the plane was left at 3,350 m.

2012: the propeler is found ~ 3 km downstream





